Empirical Proof Record: VALIDATED

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Proof ID: `46579fd975238b022e6355b50b484b62…`
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1. Claim

> Across 30 synthetic three-group datasets of size 30 per group sweeping effect size from null to strong (seed=20260518), "scipy.stats.f_oneway", "statsmodels.stats.anova.anova_lm" (via OLS), and "pingouin.anova" produce F-statistics AND two-sided p-values that agree to ≤ 1e-09 across all pairwise comparisons. (RFC 0002 Phase E1: cross-framework validation; three-way ANOVA variant.)

- **Operationalisation:** Generate N three-group datasets where each group has size n and is drawn from Normal(μ_g, σ^2) for a sweep of (μ_0, μ_1, μ_2) configurations including null (all μ equal) and alternatives. Compute F and two-sided p under scipy.stats.f_oneway, statsmodels OLS+anova_lm, and pingouin.anova. Compute max_abs_diff = max over (dataset, statistic ∈ {F, p}, backend-pair) of |stat_a - stat_b|. VALIDATED iff max_abs_diff ≤ tolerance.
- **Threshold:** `max_absolute_anova_difference <= 1e-09 F_or_p`
- **H0:** At least one backend pair disagrees on the ANOVA F statistic OR the two-sided p value by more than the floating-point tolerance.
- **H1:** All three backends compute identical F AND p to floating-point tolerance across every dataset.

2. Verdict

VALIDATED — observed `7.105427357601002e-14`
30 datasets × 3 backends × 2 statistics; max pairwise |Δ| = 7.105e-14 on (F, scipy_vs_statsmodels) (≤ 1e-09 tol)

3. Evidence

Pillar	Statistic	Val
anova_scipy_vs_statsmodels_vs_pingouin	max_absolute_anova_difference	7.105427357601002e-

4. Pre-registration

- Registered at: 2026-05-24T03:29:05.180056+00:00
- Config hash: c8e368fe8b61be3829c27bb61d9b3208…
- Data hash: c8e368fe8b61be3829c27bb61d9b3208…

5. Signature

d453e1a9d3c370b2da0b5429be6f35bf454796e84862bcbf98d65c5d78d9eefc